

- Q.26 Write locking unit and injection unit.  
 Q.27 Name various mould release agents.  
 Q.28 Explain with diagram close die press forging process.  
 Q.29 Explain with diagram close die hammer forging.  
 Q.30 Draw a neat sketch of push back type ejector return mechanism.

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (25x2=50)  
 Q.31 Explain the design procedure of an upsetting die forging for a simple component with the help of neat sketch.  
 Q.32 Explain the design consideration of fuller and finisher.  
 Q.33 Explain main parts of gravity die casting machine with neat and clean sketch.

No. of Printed Pages : 4                      181863/121863/031862  
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#### 6th Sem / T&D

#### Subject:- Forging and Casting Dies-Design & Drawing

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Write the another name for gravity die casting  
 a) Hot chambered die casting  
 b) Cold chambered die casting  
 c) Permanent Mold casting  
 d) None of the above
- Q.2 In forging process metal is used in following states.  
 a) Liquid  
 b) Solid  
 c) Both a & b  
 d) Heated up to recrystallization temperature
- Q.3 Core is made up of  
 a) Sand                                      b) Metal  
 c) Both a & b                              d) None of the above
- Q.4 Swaging is an operation of  
 a) Forging                                      b) Extrusion  
 c) Piercing                                      d) Rolling

- Q.5 In the forging operation, fullering is done to  
 a) Upset the material    b) Bend the material  
 c) Draw out the material    d) Extrude the material
- Q.6 Foundry crucible is made up of  
 a) Led                              b) Cast Iron  
 c) Graphite                      d) Mild steel
- Q.7 The forging method used to reduce the cross sectional area is called  
 a) Upsetting                      b) Edging  
 c) Heading                      d) Fullering
- Q.8 One of the advantage given bellow of the forged part is not correct  
 a) High Protection Rate  
 b) Higher Forging head  
 c) Less die wear  
 d) Lower forging temperature
- Q.9 In hammer forging, the hammer does not consist of  
 a) falling weight  
 b) Anvil  
 c) Die  
 d) Lifting mechanism of the ram
- Q.10 The property of a material necessary for forging, in stamping images on coins and in ornamental work is  
 a) Elasticity                      b) Plasticity  
 c) Ductility                      d) Malleability

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## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define fullering.
- Q.12 Define permanent mould casting.
- Q.13 Name the material used in making die casting dies.
- Q.14 Define upsetting.
- Q.15 What is cold forging.
- Q.16 Which type materials are cast in hot chamber die casting.
- Q.17 What is the another name of permanent mould casting.
- Q.18 Write the name of any two furnaces used in casting.
- Q.19 Define draft angle.
- Q.20 Write any two tools used in forging dies.

## SECTION-C

**Note:** Short answer type questions. Attempt any six questions out of ten questions. (5x6=30)

- Q.21 What is investment casting process.
- Q.22 What are the various cooling methods of pressure die casting.
- Q.23 Describe various parts of open die casting machine.
- Q.24 What are the different parts of gravity die casting?
- Q.25 Explain various types of gating systems used in casting.

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